Endothelial QC and subtypes · RNA + Xenium

RNA: 5,573 previously labeled endothelial nuclei; 4,597 in the primary reanalysis; 976 retained as weak-lineage or mixed-program review.

All RNA input nuclei already passed per-library scDblFinder and converged DecontX ambient correction. This task verifies those calls and adds endothelial-specific residual-program QC; ambient correction is not repeated on an isolated lineage.

Coherent EC/mural programs in both RNA libraries enter group review and are excluded from the final primary graph. Xenium parents with multiple projected nuclear components enter merge/multinucleation review; no successful cell splitting is asserted.

RNA discovery: independent clustering at P0, P7, P14 and P21. Within-age library effects are adjusted; raw counts, unadjusted graphs, nuisance adjustment, marker withholding, seeds, resolutions and an ambient<=30% cohort are compared. Gene-based names identify enrichment programs, with internal IDs retained in tables.

Xenium: all original broad ECs plus uncertain EC proposals are audited with the completed image-only whole-section nuclear-anchored segmentation. Native, refined cell, boundary core, nuclear-only, cytoplasmic-only and molecule-depth controls are compared. No droplet doublet or ambient algorithm is imposed on segmented profiles.

Strict A/B labels require cell and nuclear agreement; C adds boundary-stable cell labels. Additional nuclear-supported subtype candidates are reported separately, including profiles whose cell bodies remain mixed. Unresolved profiles remain in the audit and gray map background.

These are operational evidence levels, not calibrated biological accuracy. One Xenium section per age limits generalization. Original coordinates and parent atlas labels are preserved.

Assignment coverage and interpretation limits

P0: 6,394 native ECs; 660 strict A/B fine labels; 711 including C; 864 extra nuclear-only subtype candidates.

P7: 16,010 native ECs; 427 strict A/B fine labels; 541 including C; 1,295 extra nuclear-only subtype candidates.

P14: 23,060 native ECs; 691 strict A/B fine labels; 745 including C; 3,583 extra nuclear-only subtype candidates.

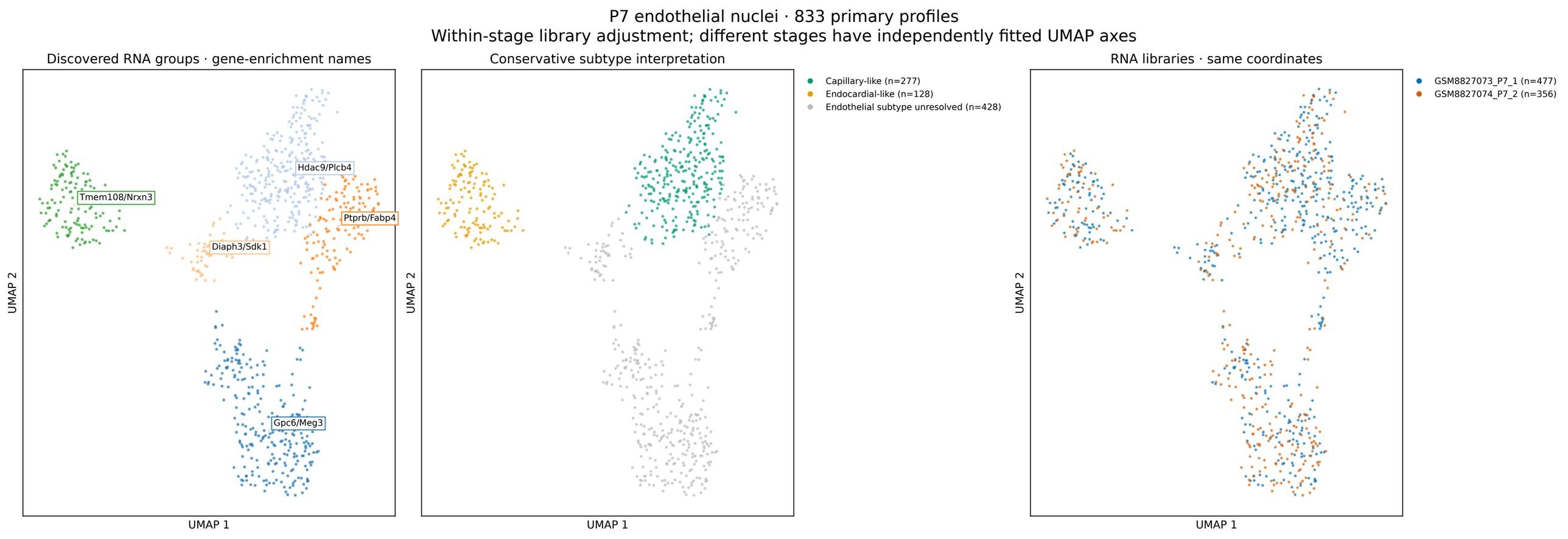
P21: 24,153 native ECs; 497 strict A/B fine labels; 549 including C; 2,524 extra nuclear-only subtype candidates.

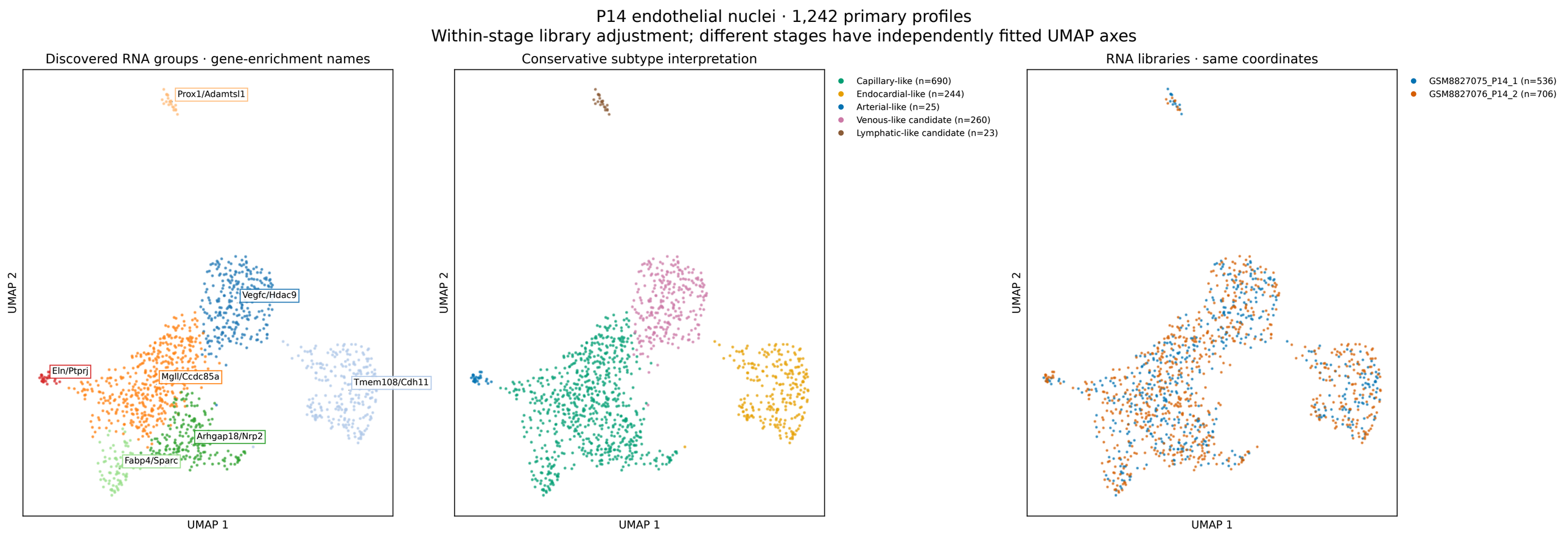
Fine labels require RNA-reference agreement with both separate libraries and raw counts, multiple measured subtype markers including an anchor, broad-lineage competition, adequate RNA information and no conflicting strong evidence. Lymphatic candidates require both Prox1 and Flt4; Lyve1 alone is insufficient.

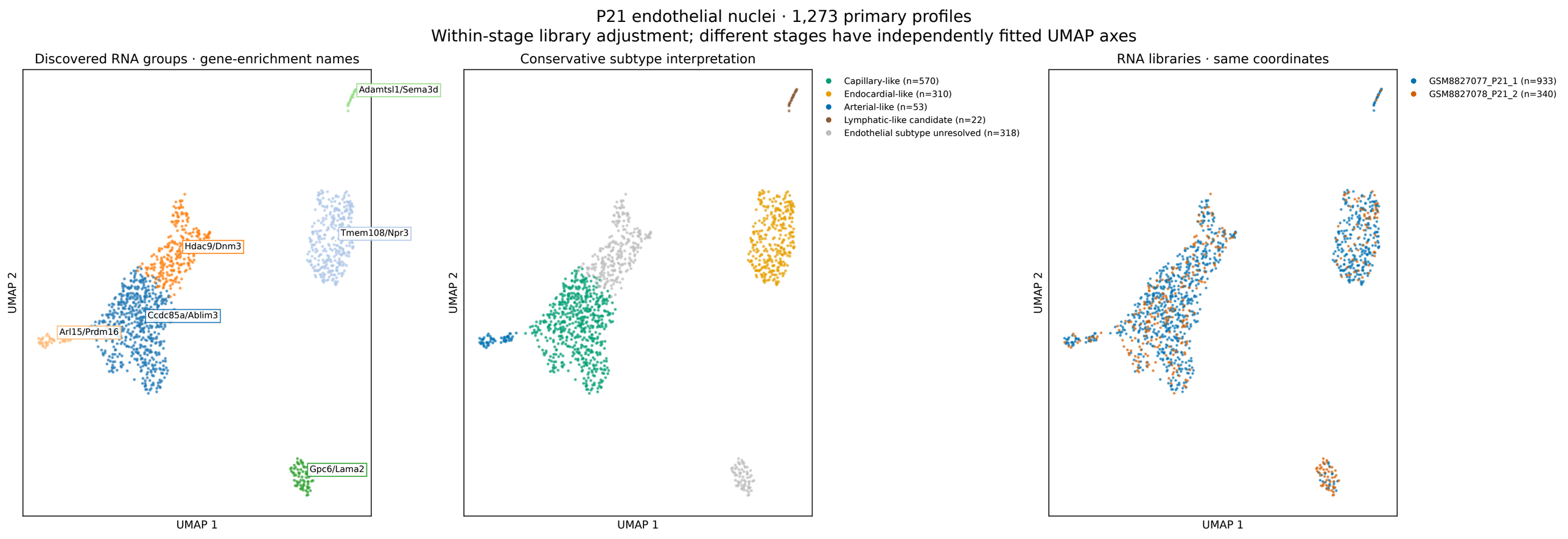
A capillary-EC population with myofiber-associated RNA was described in the source paper. Our mixed-program flags do not prove those cells are doublets or that the RNA is biologically endogenous. Nuclear persistence and boundary/depth sensitivity are reported as evidence, not a binary biological verdict.

Source: postnatal heart atlas, DOI10.1038/s44161-026-00861-z; snRNA GSE290989; Xenium Zenodo14961929 and14961942. Paper Supplementary Table3 markers supplement curated EC panels. Full methods, marker evidence, counts and validation accompany this report.

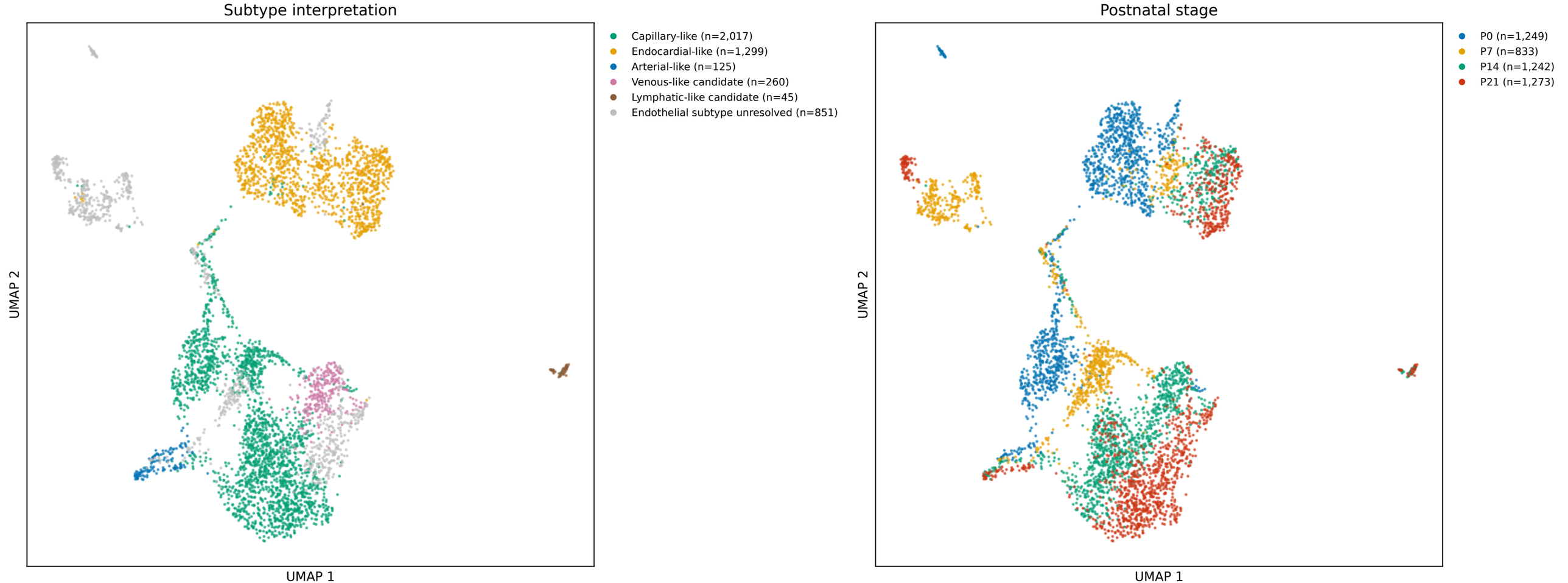

P0 endothelial nuclei · 1,249 primary profiles
Within-stage library adjustment; different stages have independently fitted UMAP axes
Discovered RNA groups · gene-enrichment names
Vegfc/Pcsk5
Tmem108/Igfbp5
Dach1/Cd36
Tmem108/Cfh
Diaph3/Top2a
Sdk1/Nckap5
UMAP 2
UMAP 1
Conservative subtype interpretation
Capillary-like (n=480)
Endocardial-like (n=617)
Arterial-like (n=47)
Endothelial subtype unresolved (n=105)
UMAP 2
UMAP 1
RNA libraries · same coordinates
GSM8827071_P0_1 (n=608)
GSM8827072_P0_2 (n=641)
UMAP 2
UMAP 1

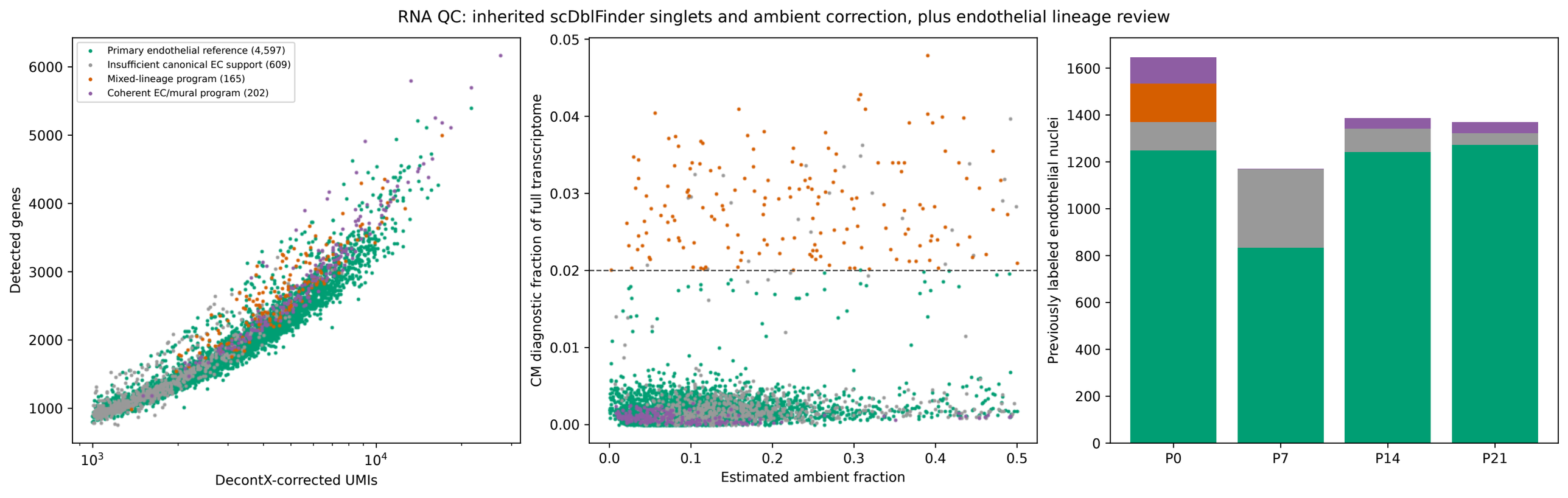

P7 endothelial nuclei · 833 primary profiles
Within-stage library adjustment; different stages have independently fitted UMAP axes
Discovered RNA groups · gene-enrichment names
Tmem108/Nrxn3
Hdac9/Plcb4
Ptprb/Fabp4
Diaph3/Sdk1
Gpc6/Meg3
UMAP 2
UMAP 1
Conservative subtype interpretation
Capillary-like (n=277)
Endocardial-like (n=128)
Endothelial subtype unresolved (n=428)
UMAP 2
UMAP 1
RNA libraries · same coordinates
GSM8827073_P7_1 (n=477)
GSM8827074_P7_2 (n=356)
UMAP 2
UMAP 1


P14 endothelial nuclei · 1,242 primary profiles
Within-stage library adjustment; different stages have independently fitted UMAP axes
Discovered RNA groups · gene-enrichment names
Prox1/Adamtsl1
Vegfc/Hdac9
Eln/Ptprj
Mgll/Ccdc85a
Tmem108/Cdh11
Arhgap18/Nrp2
Fabp4/Sparc
UMAP 2
UMAP 1
Conservative subtype interpretation
Capillary-like (n=690)
Endocardial-like (n=244)
Arterial-like (n=25)
Venous-like candidate (n=260)
Lymphatic-like candidate (n=23)
UMAP 2
UMAP 1
RNA libraries · same coordinates
GSM8827075_P14_1 (n=536)
GSM8827076_P14_2 (n=706)
UMAP 2
UMAP 1


P21 endothelial nuclei · 1,273 primary profiles
Within-stage library adjustment; different stages have independently fitted UMAP axes
Discovered RNA groups · gene-enrichment names
Adamtsl1/Sema3d
Tmem108/Npr3
Hdac9/Dnm3
Ccdc85a/Ablim3
Arl15/Prdm16
Gpc6/Lama2
UMAP 2
UMAP 1
Conservative subtype interpretation
Capillary-like (n=570)
Endocardial-like (n=310)
Arterial-like (n=53)
Lymphatic-like candidate (n=22)
Endothelial subtype unresolved (n=318)
UMAP 2
UMAP 1
RNA libraries · same coordinates
GSM8827077_P21_1 (n=933)
GSM8827078_P21_2 (n=340)
UMAP 2
UMAP 1

All-stage endothelial RNA · same UMAP, no batch/age integration
Stage and library effects can both contribute to separation



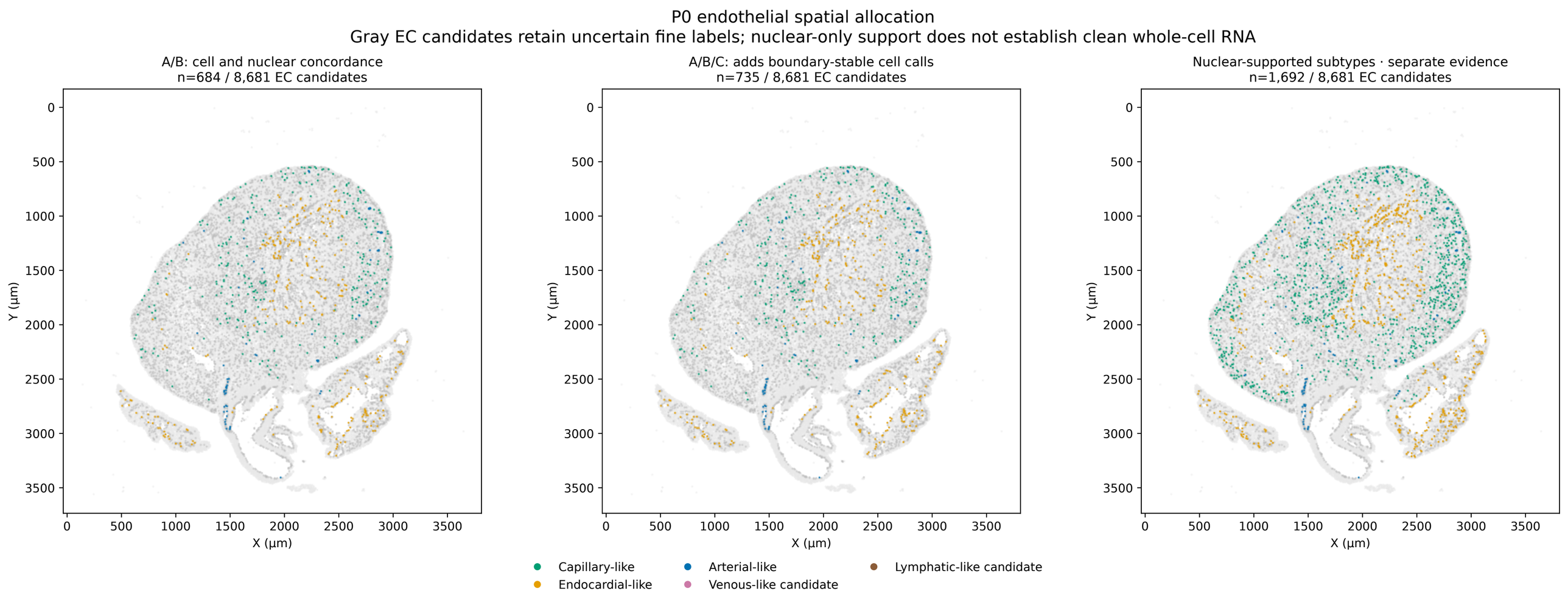

RNA QC: inherited scDblFinder singlets and ambient correction, plus endothelial lineage review
Primary endothelial reference (4,597)
Insufficient canonical EC support (609)
Mixed-lineage program (165)
Coherent EC/mural program (202)
Detected genes
DecontX-corrected UMIs
CM diagnostic fraction of full transcriptome
Estimated ambient fraction
Previously labeled endothelial nuclei
P0
P7
P14
P21

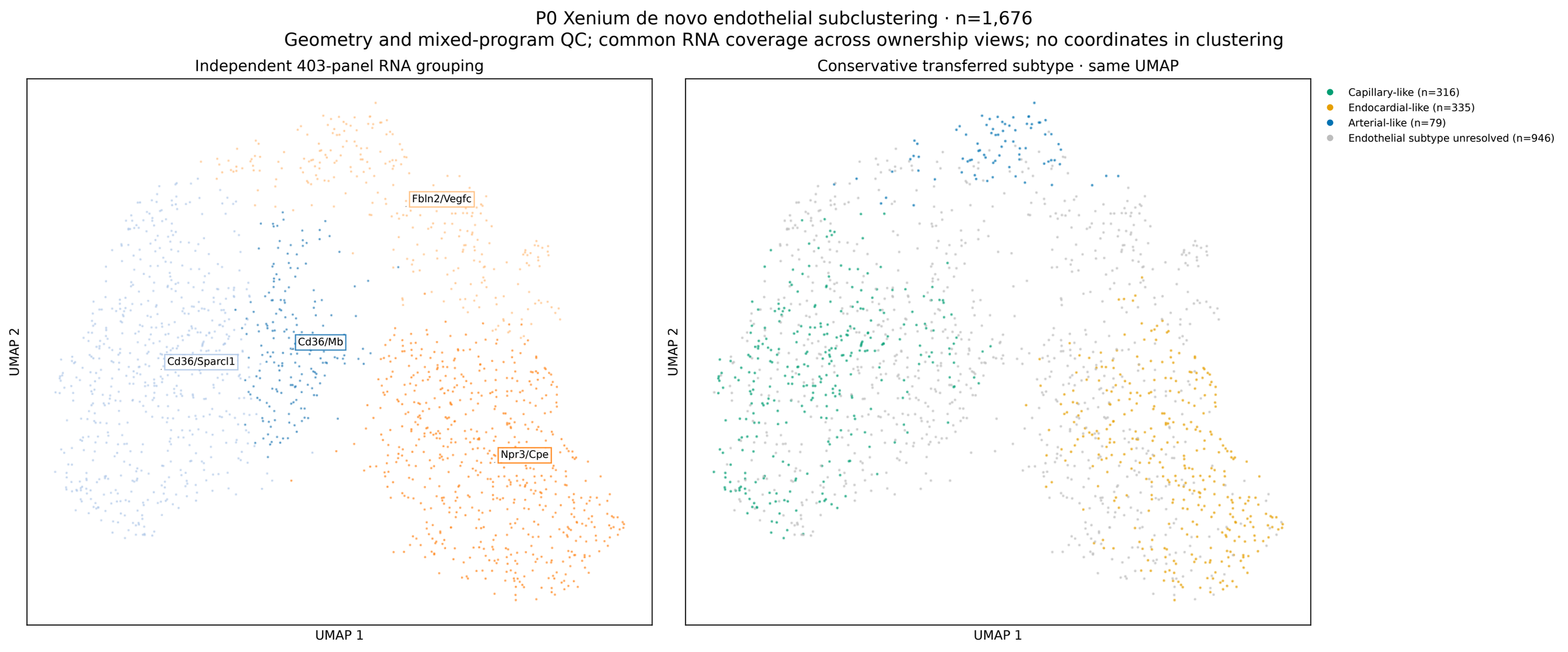
Endothelial marker evidence · color: mean log1p CP10k; dot area: detection fraction

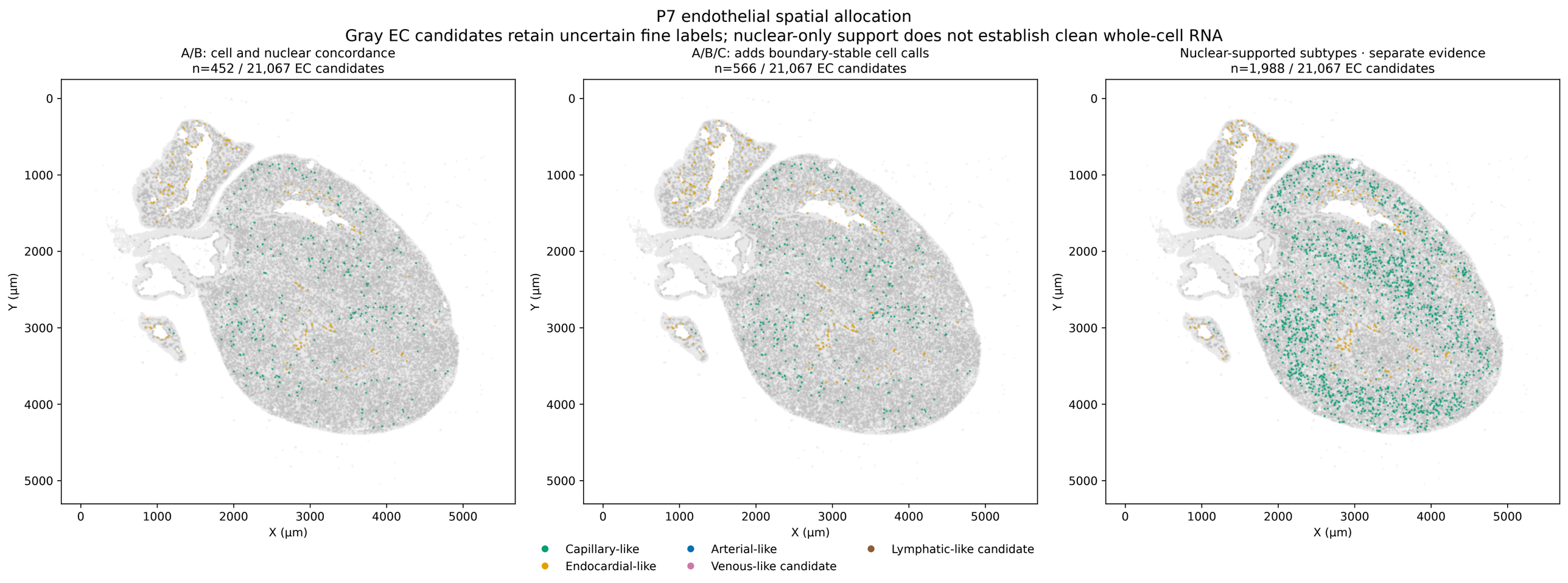
P0 · Dach1/Cd36 ECs
P0 · Tmem108/Cfh ECs
P0 · Sdk1/Nckap5 ECs
P0 · Tmem108/Igfbp5 ECs
P0 · Vegfc/Pcsk5 ECs
P0 · Diaph3/Top2a ECs
P7 · Gpc6/Meg3 ECs
P7 · Hdac9/Plcb4 ECs
P7 · Ptprb/Fabp4 ECs
P7 · Diaph3/Sdk1 ECs
P7 · Tmem108/Nrxn3 ECs
P14 · Vegfc/Hdac9 ECs
P14 · Tmem108/Cdh11 ECs
P14 · Mgll/Ccdc85a ECs
P14 · Prox1/Adamtsl1 ECs
P14 · Arhgap18/Nrp2 ECs
P14 · Fabp4/Sparc ECs
P14 · Eln/Ptprj ECs
P21 · Ccdc85a/Ablim3 ECs
P21 · Tmem108/Npr3 ECs
P21 · Hdac9/Dnm3 ECs
P21 · Arl15/Prdm16 ECs
P21 · Gpc6/Lama2 ECs
P21 · Adamtsl1/Sema3d ECs

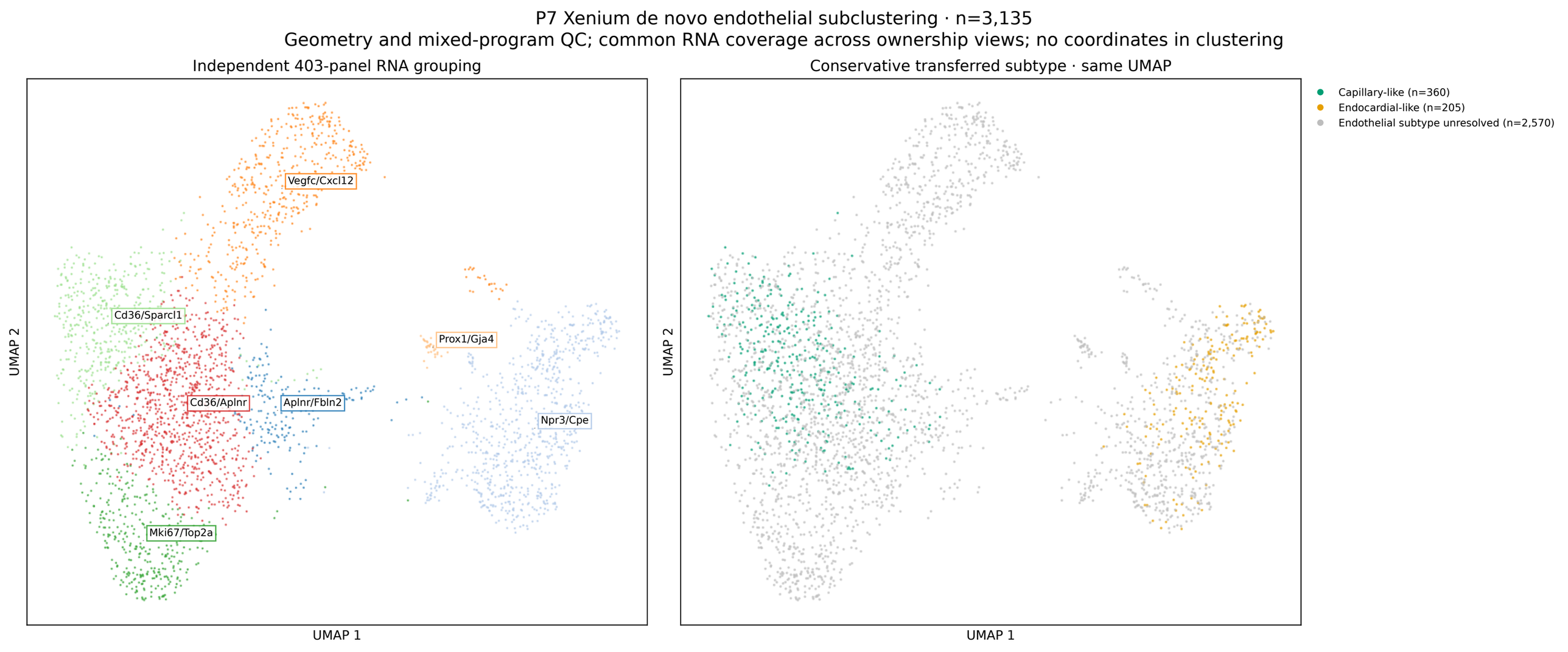
Cdh5
Pecam1
Erg
Dach1
Cd36
Rgcc
Emcn
Npr3
Nrxn3
Nfatc1
Gja5
Sox17
Sema3g
Ephb4
Nr2f2
Prox1
Flt4
Ccl21a
Top2a
Mki67

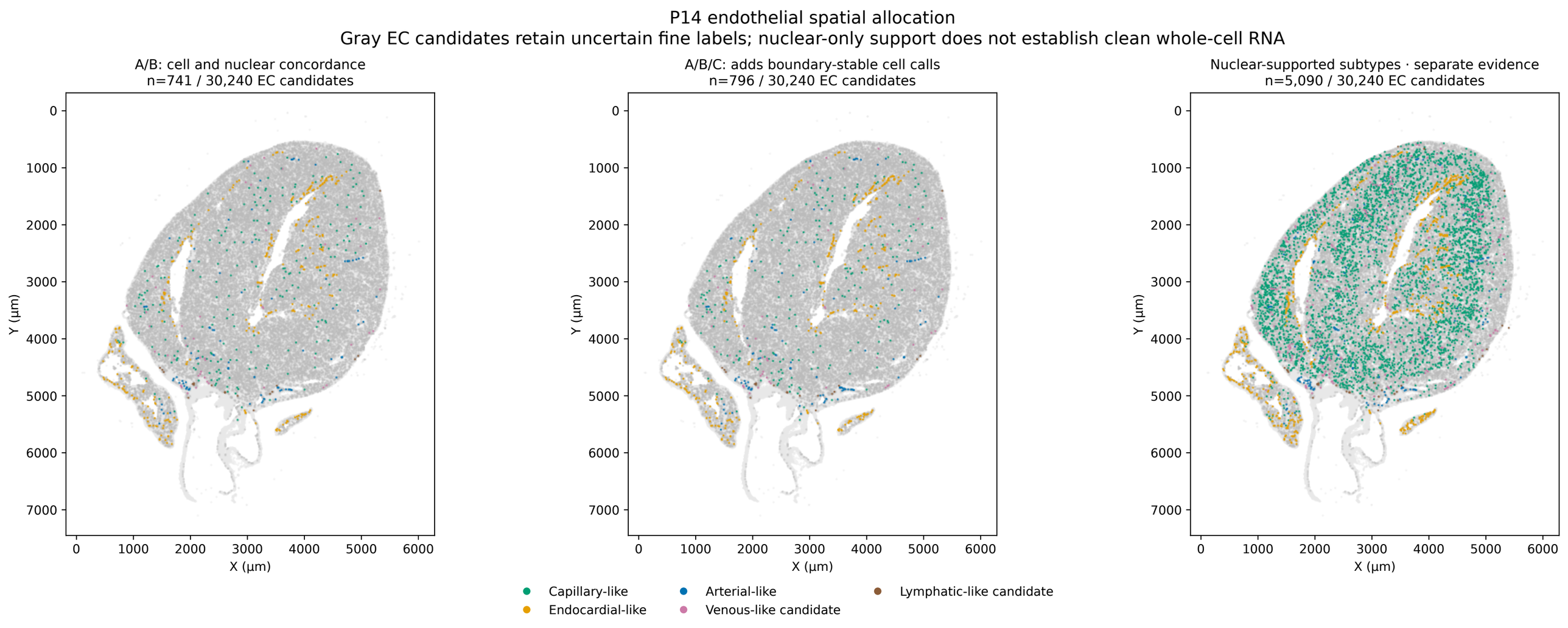
Mean log1p CP10k
3.0
2.5
2.0
1.5
1.0
0.5
0.0

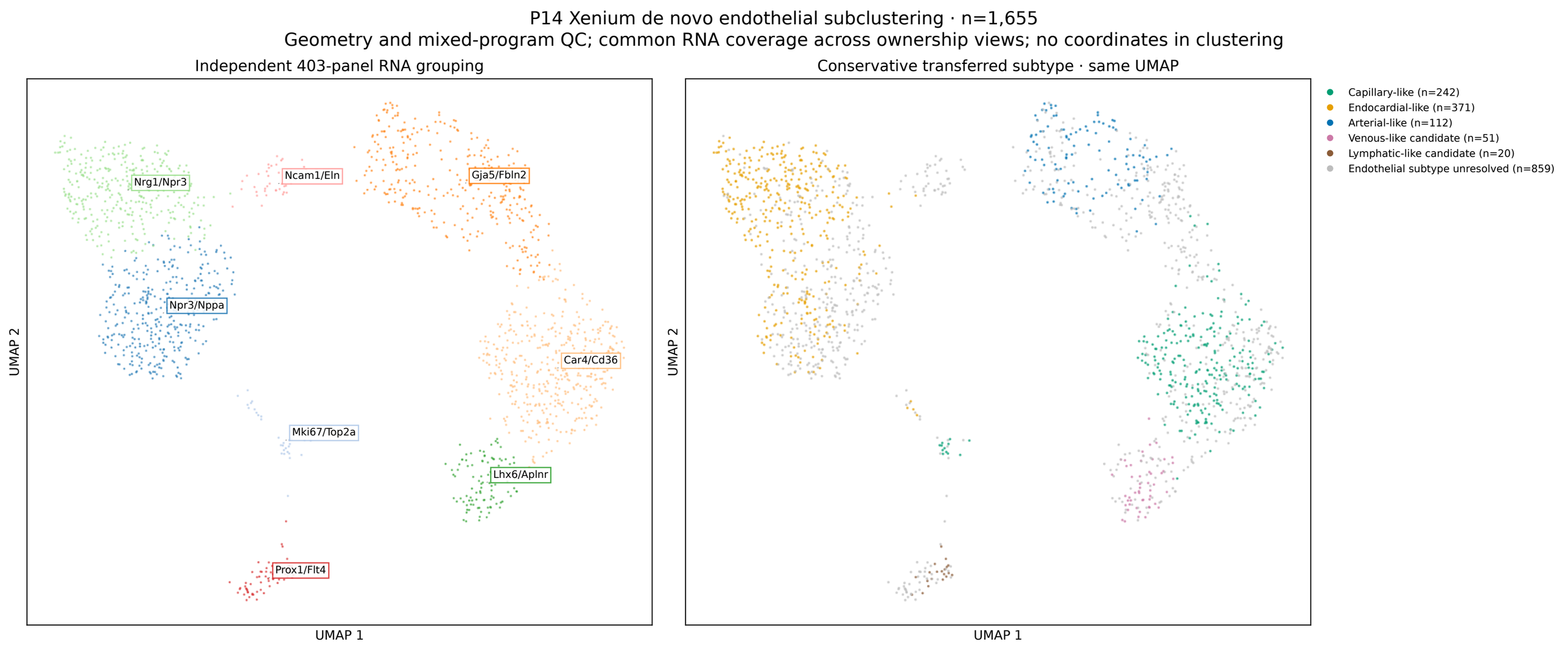

P0 endothelial spatial allocation
Gray EC candidates retain uncertain fine labels; nuclear-only support does not establish clean whole-cell RNA
A/B: cell and nuclear concordance
n=684 / 8,681 EC candidates
A/B/C: adds boundary-stable cell calls
n=735 / 8,681 EC candidates
Nuclear-supported subtypes · separate evidence
n=1,692 / 8,681 EC candidates
Y (µm)
X (µm)
0
500
1000
1500
2000
2500
3000
3500
Capillary-like
Endocardial-like
Arterial-like
Venous-like candidate
Lymphatic-like candidate

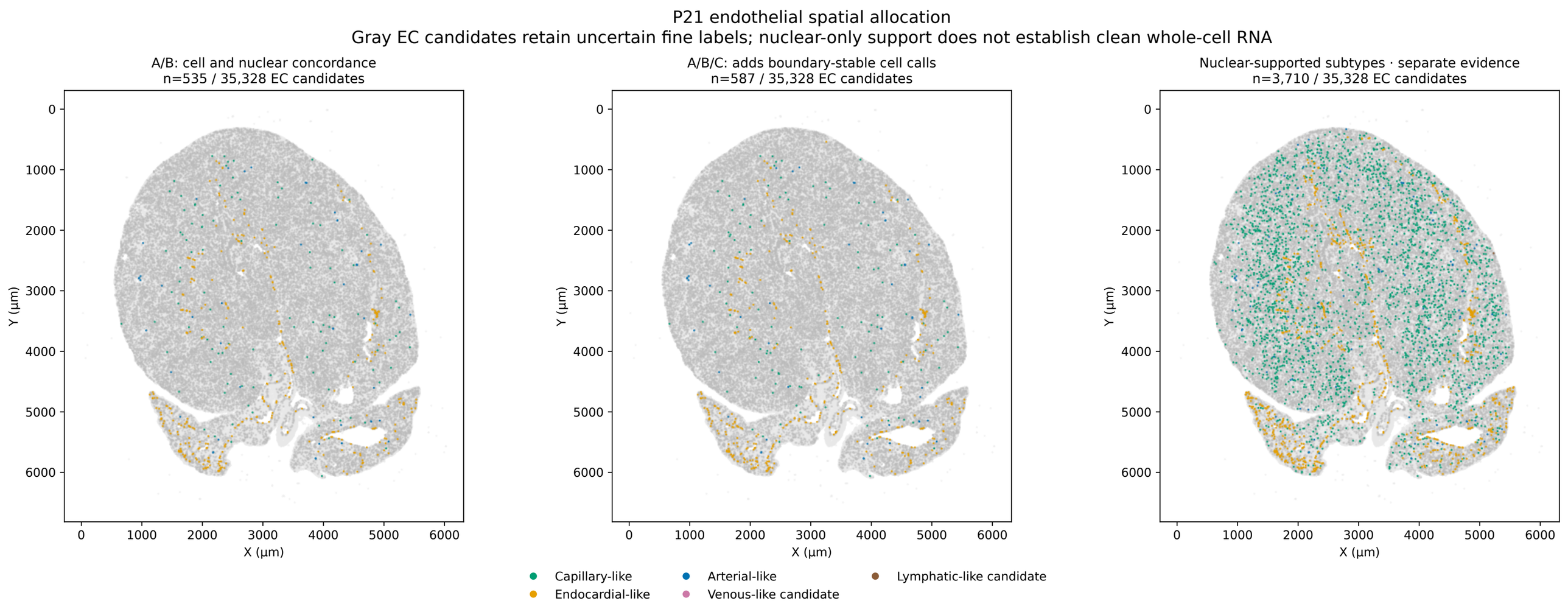

P0 Xenium de novo endothelial subclustering · n=1,676
Geometry and mixed-program QC; common RNA coverage across ownership views; no coordinates in clustering
Independent 403-panel RNA grouping
Fbln2/Vegfc
Cd36/Mb
Cd36/Sparcl1
Npr3/Cpe
UMAP 2
UMAP 1
Conservative transferred subtype · same UMAP
Capillary-like (n=316)
Endocardial-like (n=335)
Arterial-like (n=79)
Endothelial subtype unresolved (n=946)
UMAP 2
UMAP 1

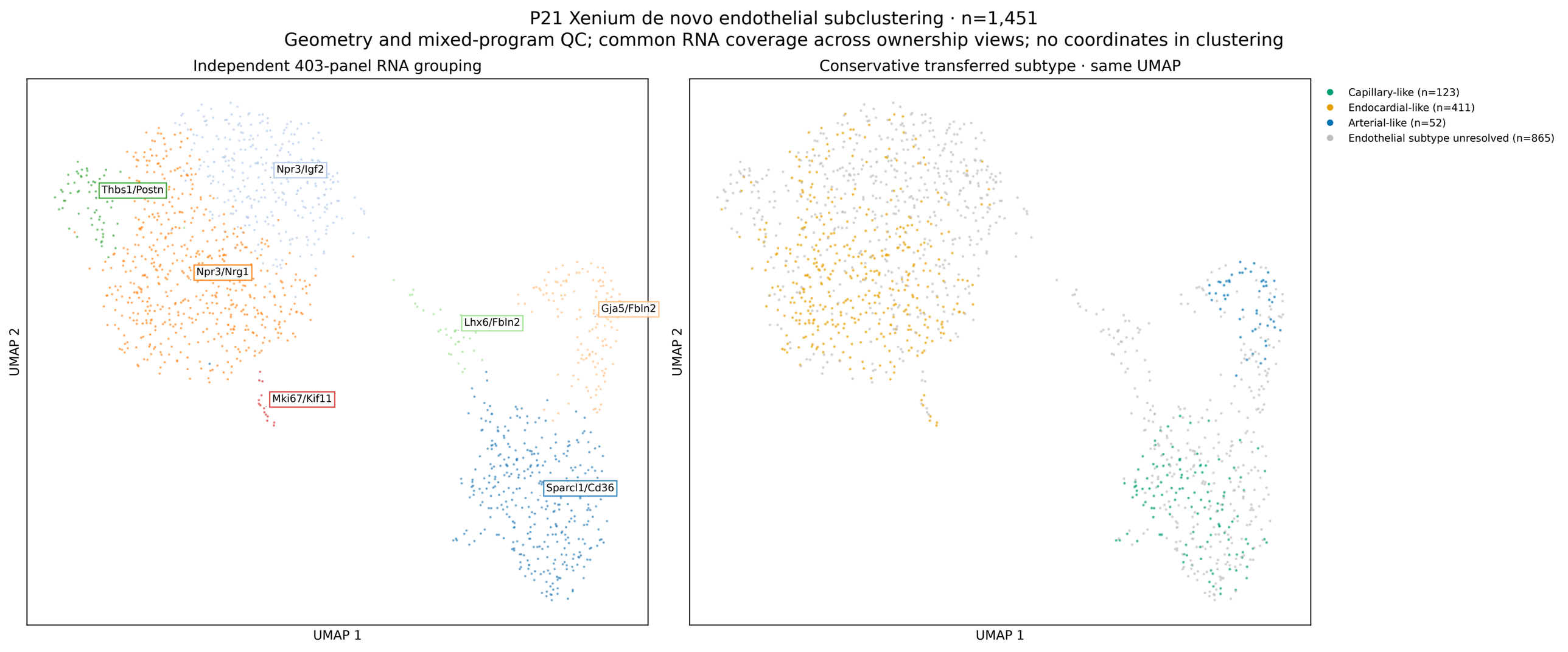

P7 endothelial spatial allocation
Gray EC candidates retain uncertain fine labels; nuclear-only support does not establish clean whole-cell RNA
A/B: cell and nuclear concordance
n=452 / 21,067 EC candidates
A/B/C: adds boundary-stable cell calls
n=566 / 21,067 EC candidates
Nuclear-supported subtypes · separate evidence
n=1,988 / 21,067 EC candidates
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
Capillary-like
Endocardial-like
Arterial-like
Venous-like candidate
Lymphatic-like candidate

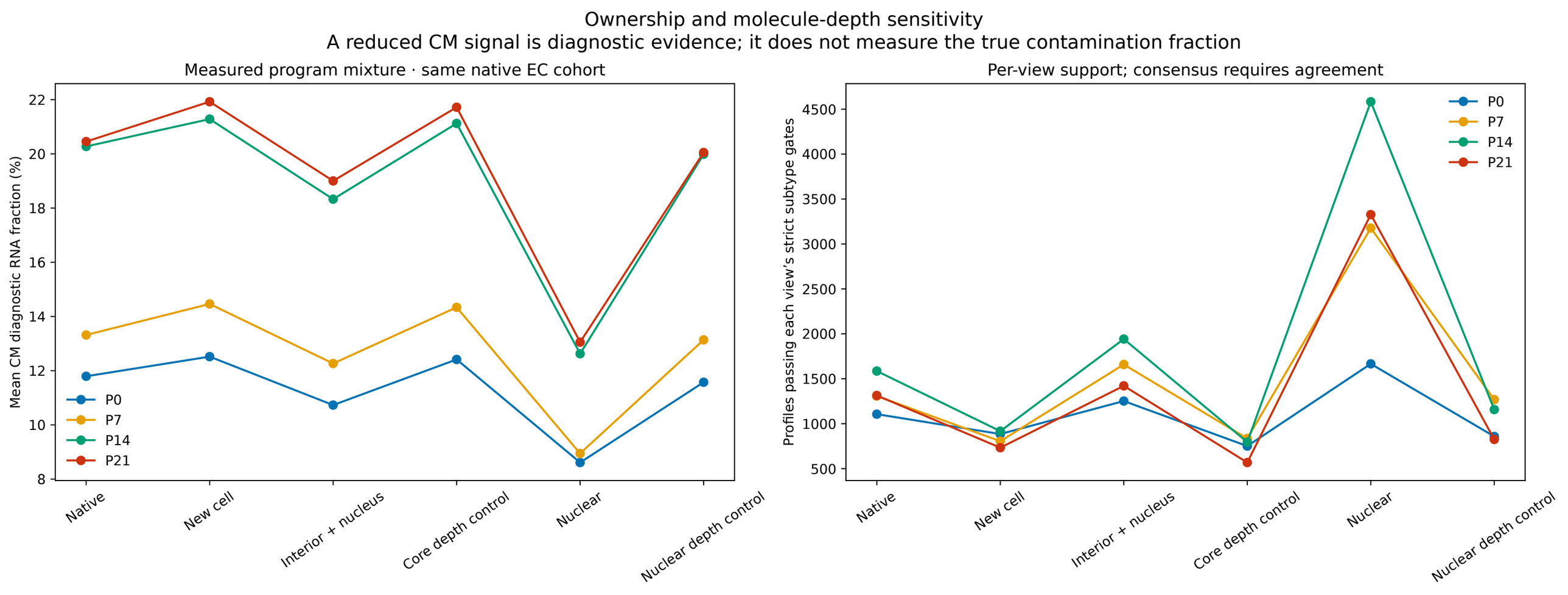

P7 Xenium de novo endothelial subclustering · n=3,135
Geometry and mixed-program QC; common RNA coverage across ownership views; no coordinates in clustering
Independent 403-panel RNA grouping
Vegfc/Cxcl12
Cd36/Sparcl1
Prox1/Gja4
Cd36/Aplnr
Aplnr/Fbln2
Npr3/Cpe
Mki67/Top2a
UMAP 2
UMAP 1
Conservative transferred subtype · same UMAP
Capillary-like (n=360)
Endocardial-like (n=205)
Endothelial subtype unresolved (n=2,570)
UMAP 2
UMAP 1

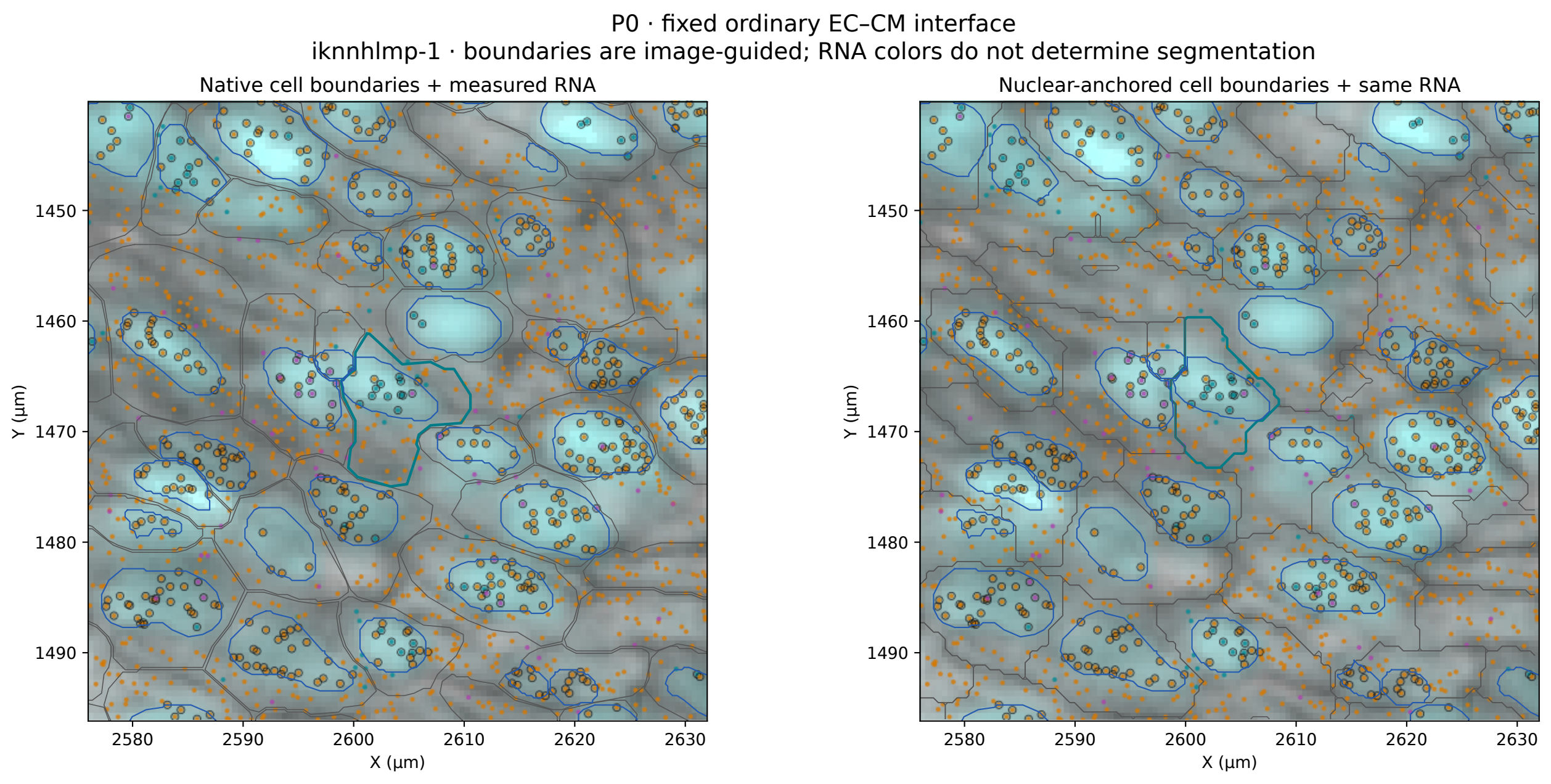

P14 endothelial spatial allocation
Gray EC candidates retain uncertain fine labels; nuclear-only support does not establish clean whole-cell RNA
A/B: cell and nuclear concordance
n=741 / 30,240 EC candidates
A/B/C: adds boundary-stable cell calls
n=796 / 30,240 EC candidates
Nuclear-supported subtypes · separate evidence
n=5,090 / 30,240 EC candidates
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
7000
Capillary-like
Endocardial-like
Arterial-like
Venous-like candidate
Lymphatic-like candidate

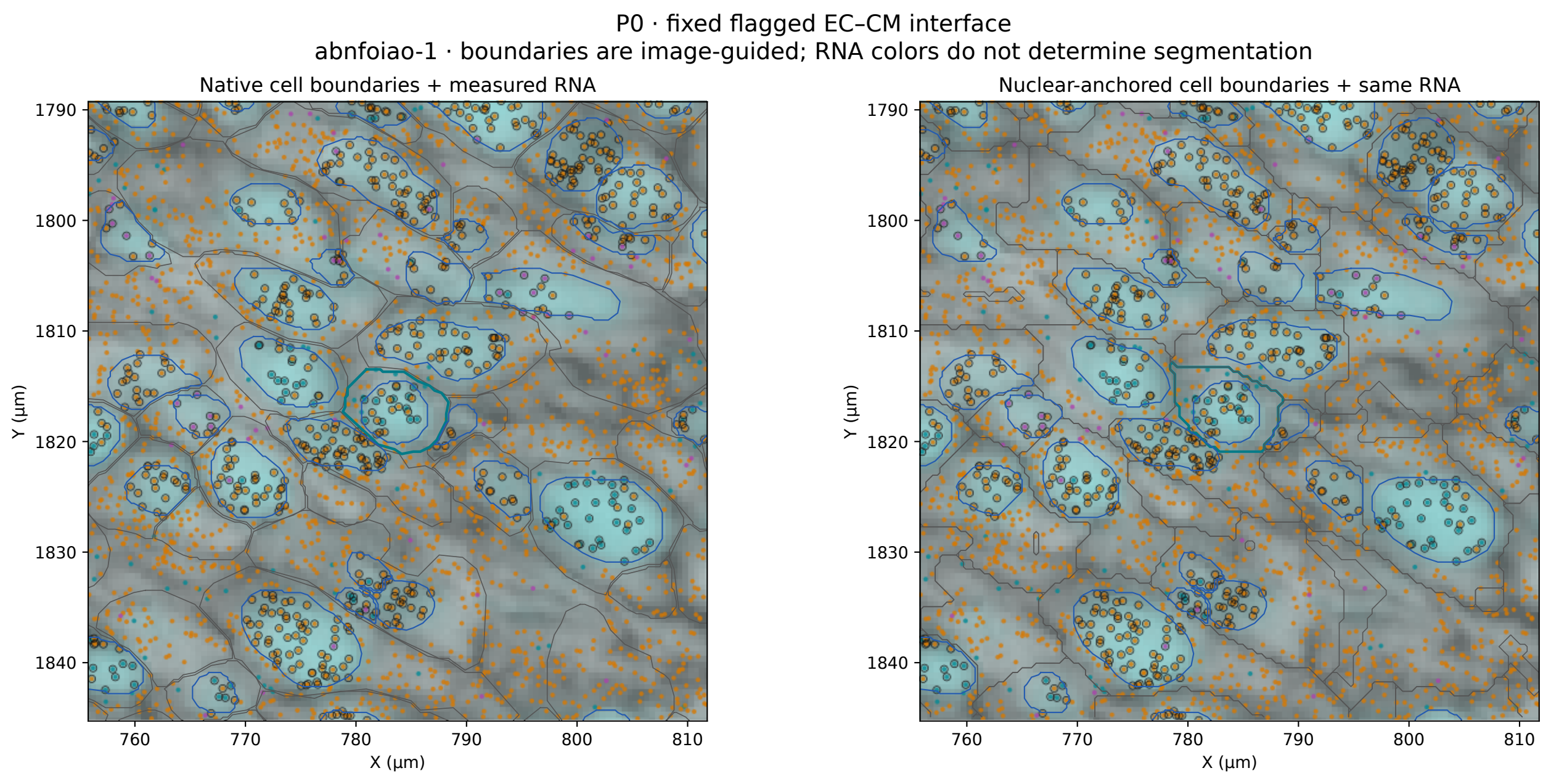

P14 Xenium de novo endothelial subclustering · n=1,655
Geometry and mixed-program QC; common RNA coverage across ownership views; no coordinates in clustering
Independent 403-panel RNA grouping
Conservative transferred subtype · same UMAP
Nrg1/Npr3
Ncam1/Eln
Gja5/Fbln2
Npr3/Nppa
Car4/Cd36
Mki67/Top2a
Lhx6/Aplnr
Prox1/Flt4
UMAP 1
UMAP 2
Capillary-like (n=242)
Endocardial-like (n=371)
Arterial-like (n=112)
Venous-like candidate (n=51)
Lymphatic-like candidate (n=20)
Endothelial subtype unresolved (n=859)

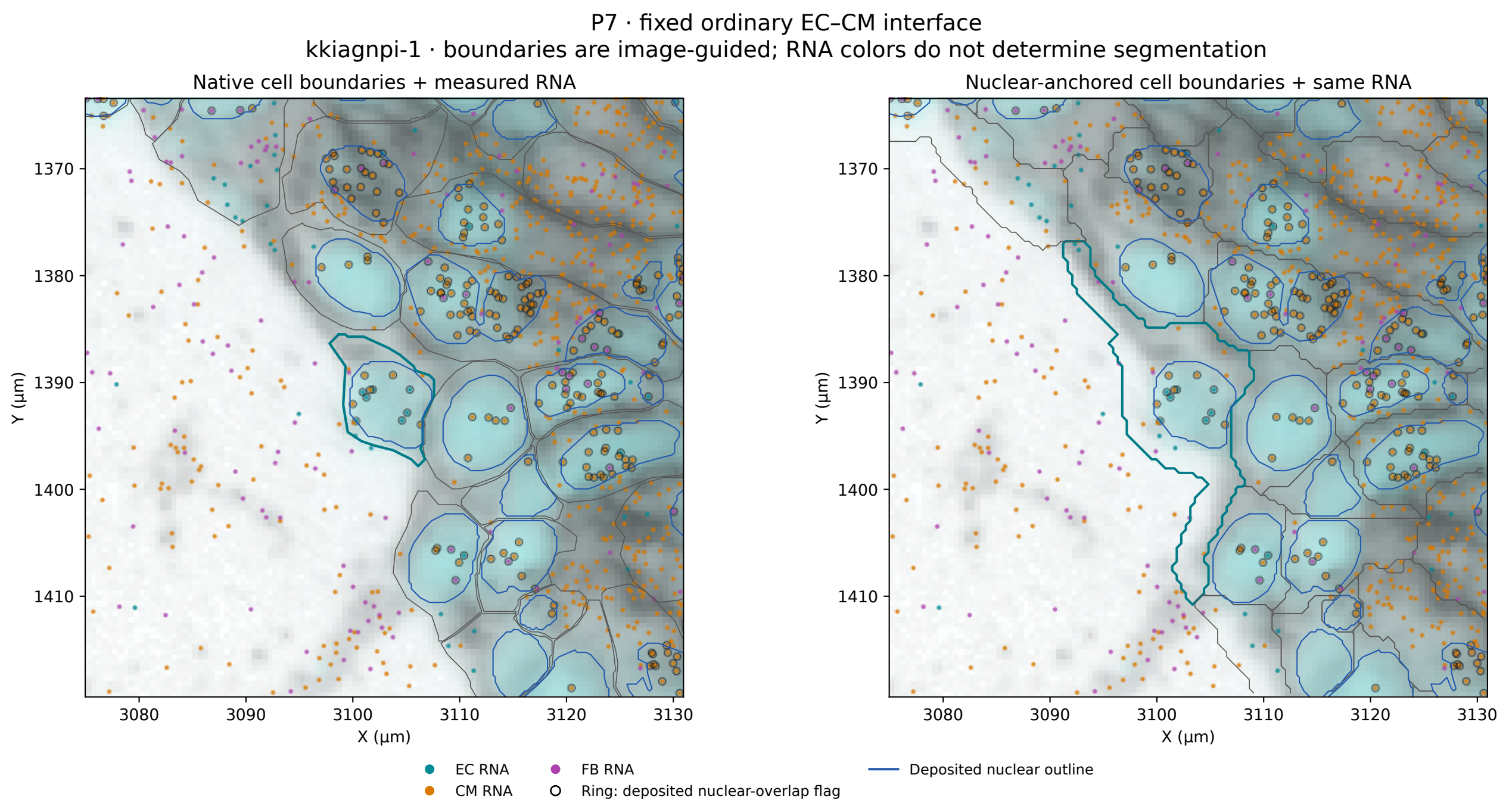

P21 endothelial spatial allocation
Gray EC candidates retain uncertain fine labels; nuclear-only support does not establish clean whole-cell RNA
A/B: cell and nuclear concordance
n=535 / 35,328 EC candidates
A/B/C: adds boundary-stable cell calls
n=587 / 35,328 EC candidates
Nuclear-supported subtypes · separate evidence
n=3,710 / 35,328 EC candidates
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
Capillary-like
Arterial-like
Lymphatic-like candidate
Endocardial-like
Venous-like candidate

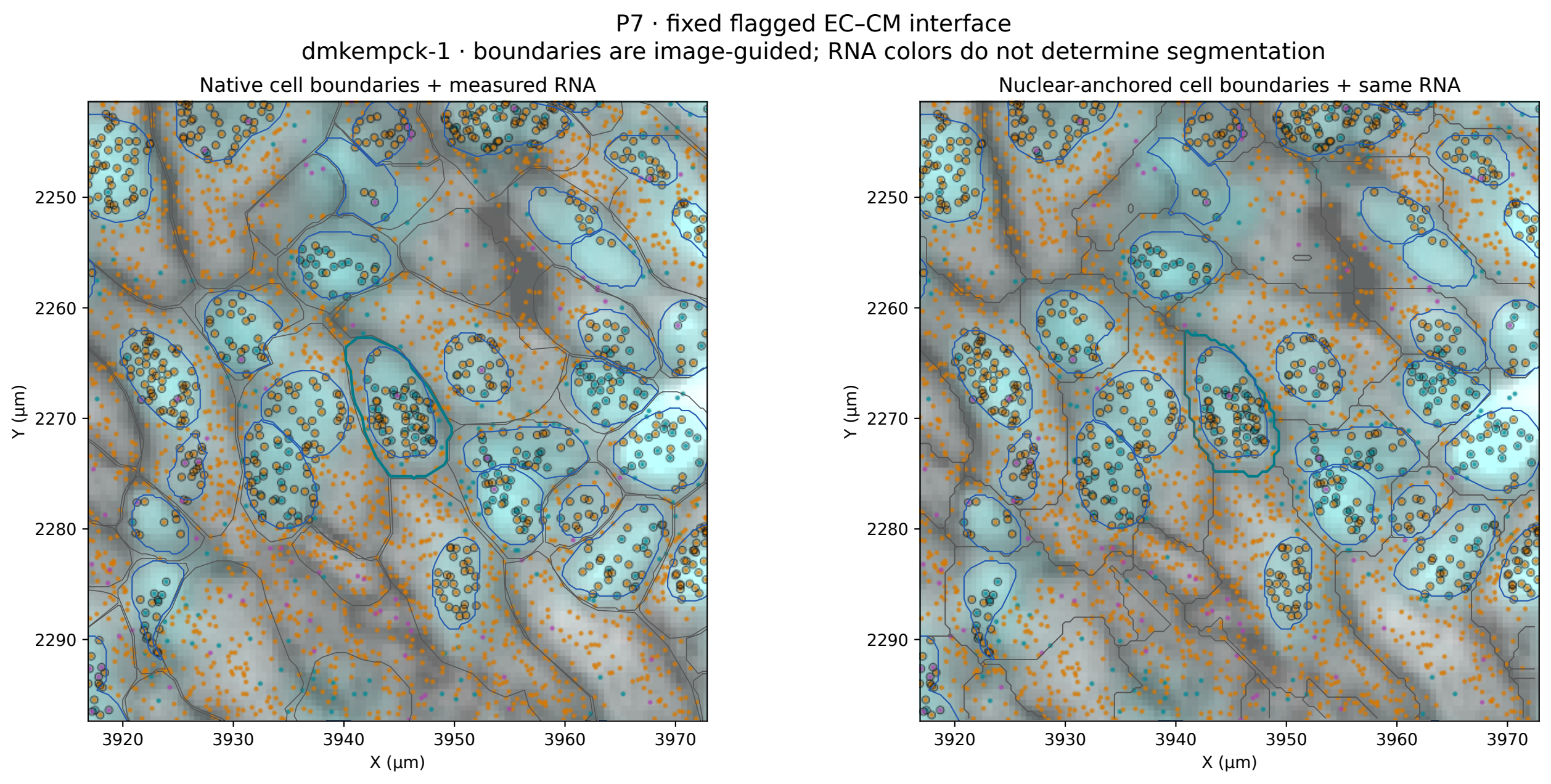

P21 Xenium de novo endothelial subclustering · n=1,451
Geometry and mixed-program QC; common RNA coverage across ownership views; no coordinates in clustering
Independent 403-panel RNA grouping
Conservative transferred subtype · same UMAP
Thbs1/Postn
Npr3/Igf2
Npr3/Nrg1
Mki67/Kif11
Lhx6/Fbln2
Gja5/Fbln2
Sparcl1/Cd36
UMAP 1
UMAP 2
UMAP 1
UMAP 2
Capillary-like (n=123)
Endocardial-like (n=411)
Arterial-like (n=52)
Endothelial subtype unresolved (n=865)

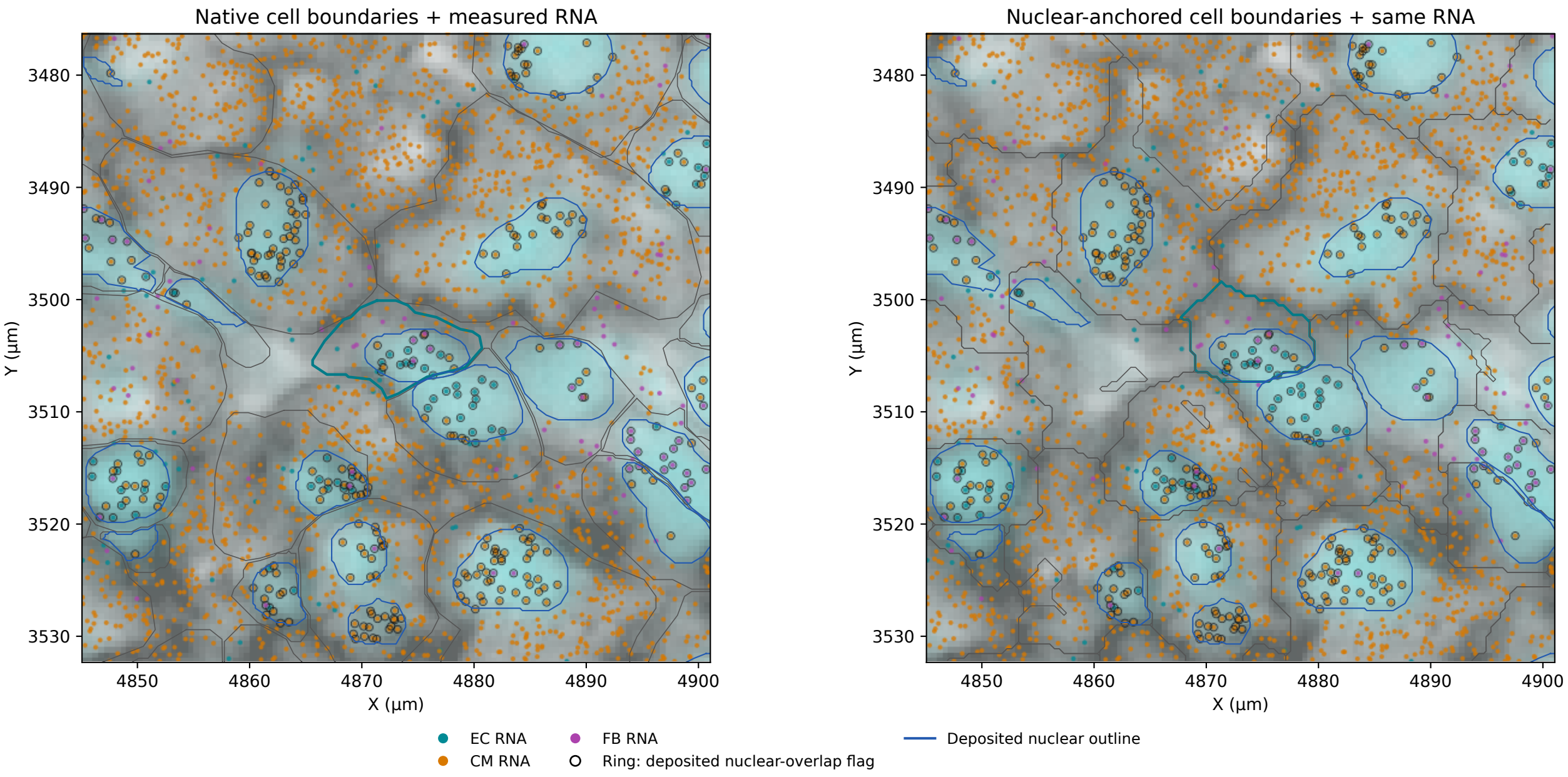

Ownership and molecule-depth sensitivity
A reduced CM signal is diagnostic evidence; it does not measure the true contamination fraction
Measured program mixture · same native EC cohort
Mean CM diagnostic RNA fraction (%)
22
20
18
16
14
12
10
8
P0
P7
P14
P21
Native
New cell
Interior + nucleus
Core depth control
Nuclear
Nuclear depth control
Per-view support; consensus requires agreement
Profiles passing each view's strict subtype gates
4500
4000
3500
3000
2500
2000
1500
1000
500
P0
P7
P14
P21
Native
New cell
Interior + nucleus
Core depth control
Nuclear
Nuclear depth control

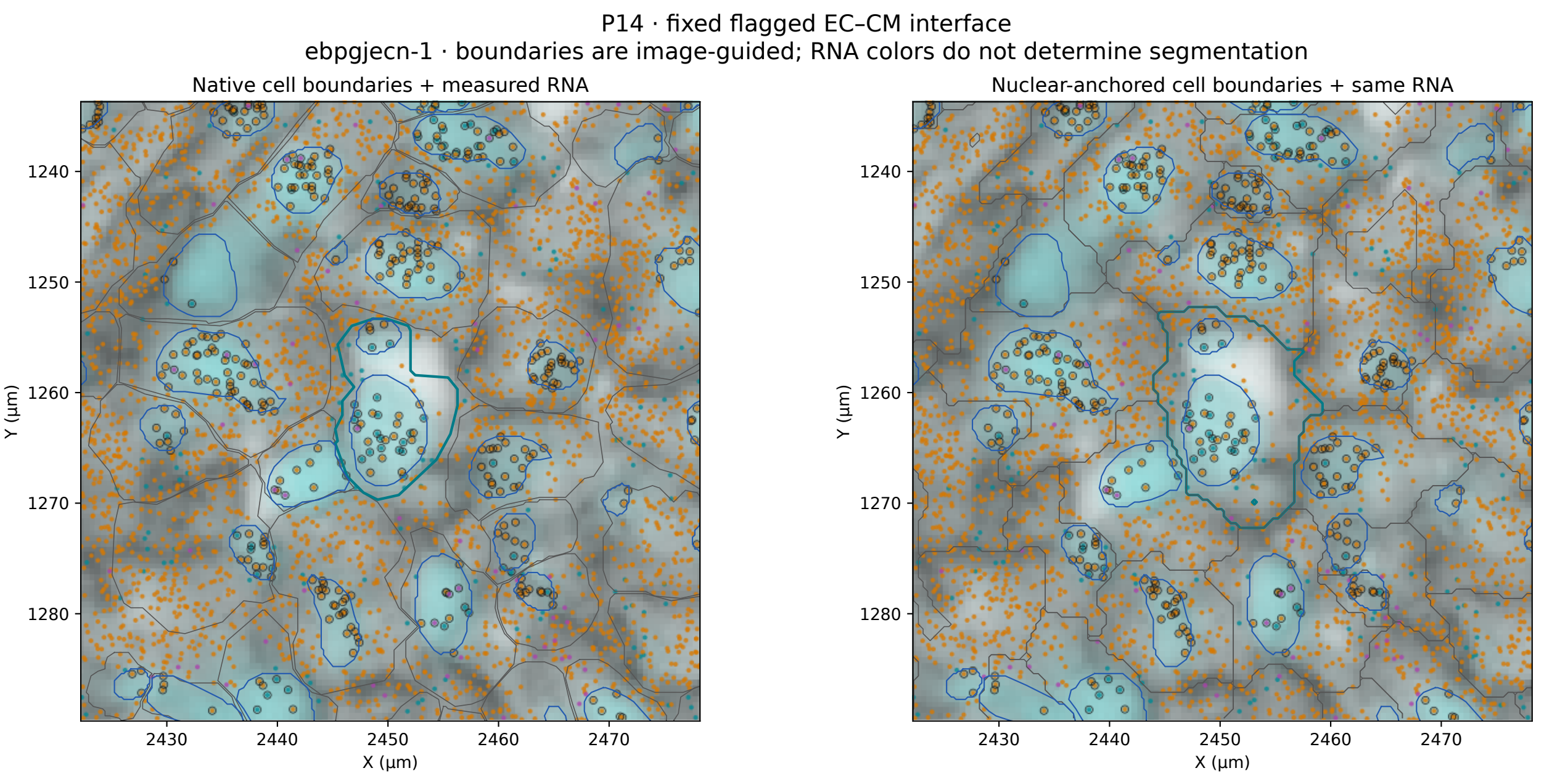

P0 · fixed ordinary EC–CM interface
iknnhlmp-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
1450
1460
1470
1480
1490
2580
2590
2600
2610
2620
2630
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline

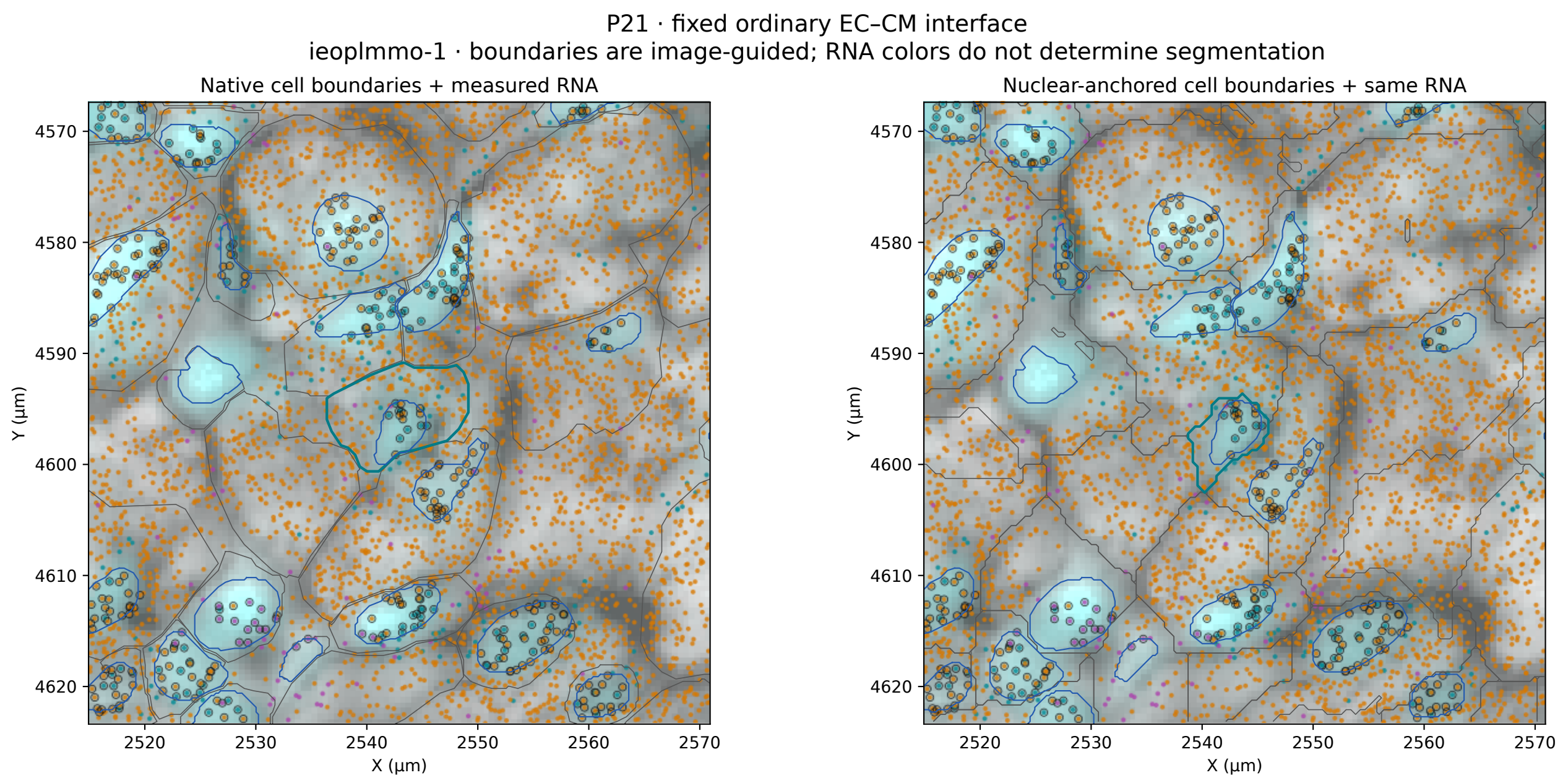

P0 · fixed flagged EC–CM interface
abnfoiao-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
1790
1800
1810
1820
1830
1840
760
770
780
790
800
810
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline

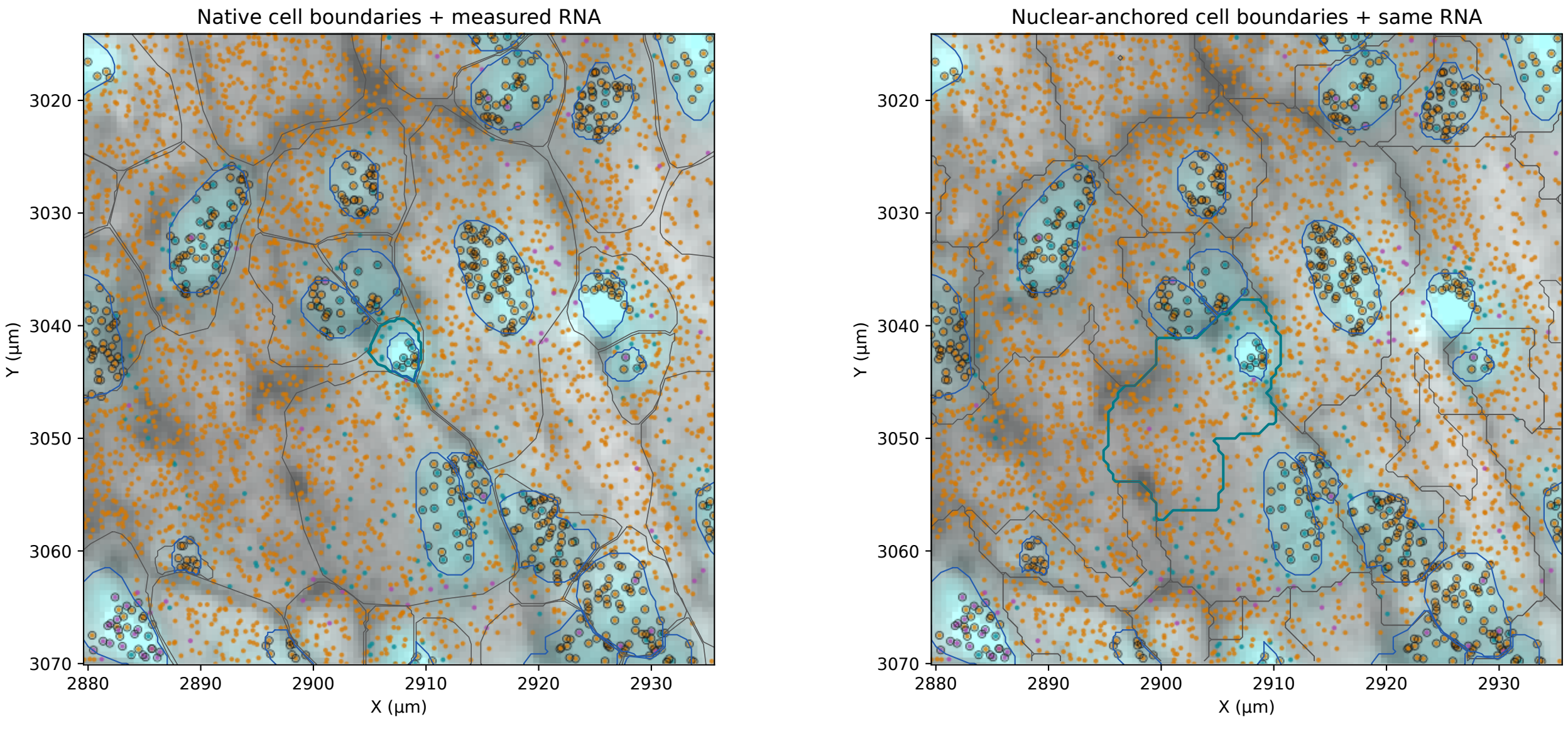

P7 · fixed ordinary EC–CM interface
kkiagnpi-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
1370
1380
1390
1400
1410
3080
3090
3100
3110
3120
3130
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline

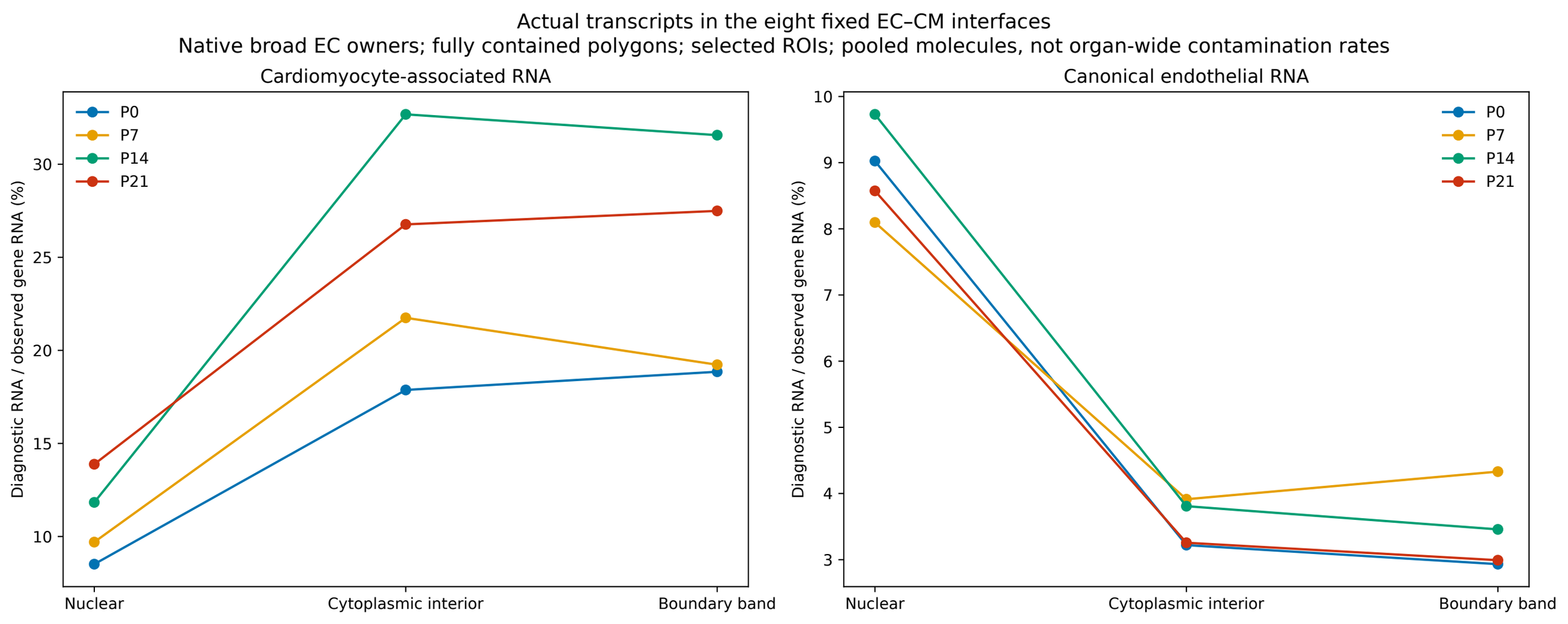

P7 · fixed flagged EC–CM interface
dmkempck-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
2250
2260
2270
2280
2290
3920
3930
3940
3950
3960
3970
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline


P14 · fixed ordinary EC–CM interface
mnolakpo-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
3480
3490
3500
3510
3520
3530
4850
4860
4870
4880
4890
4900
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline


P14 · fixed flagged EC–CM interface
ebpgjecn-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
1240
1250
1260
1270
1280
2430
2440
2450
2460
2470
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline


P21 · fixed ordinary EC–CM interface
ieoplmmo-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
4570
4580
4590
4600
4610
4620
2520
2530
2540
2550
2560
2570
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline


P21 · fixed flagged EC–CM interface
jhmmjcpa-1 · boundaries are image-guided; RNA colors do not determine segmentation
Native cell boundaries + measured RNA
Nuclear-anchored cell boundaries + same RNA
Y (µm)
X (µm)
3020
3030
3040
3050
3060
3070
2880
2890
2900
2910
2920
2930
EC RNA
CM RNA
FB RNA
Ring: deposited nuclear-overlap flag
Deposited nuclear outline


Actual transcripts in the eight fixed EC–CM interfaces
Native broad EC owners; fully contained polygons; selected ROIs; pooled molecules, not organ-wide contamination rates
Cardiomyocyte-associated RNA
Canonical endothelial RNA
Diagnostic RNA / observed gene RNA (%)
P0
P7
P14
P21
Nuclear
Cytoplasmic interior
Boundary band